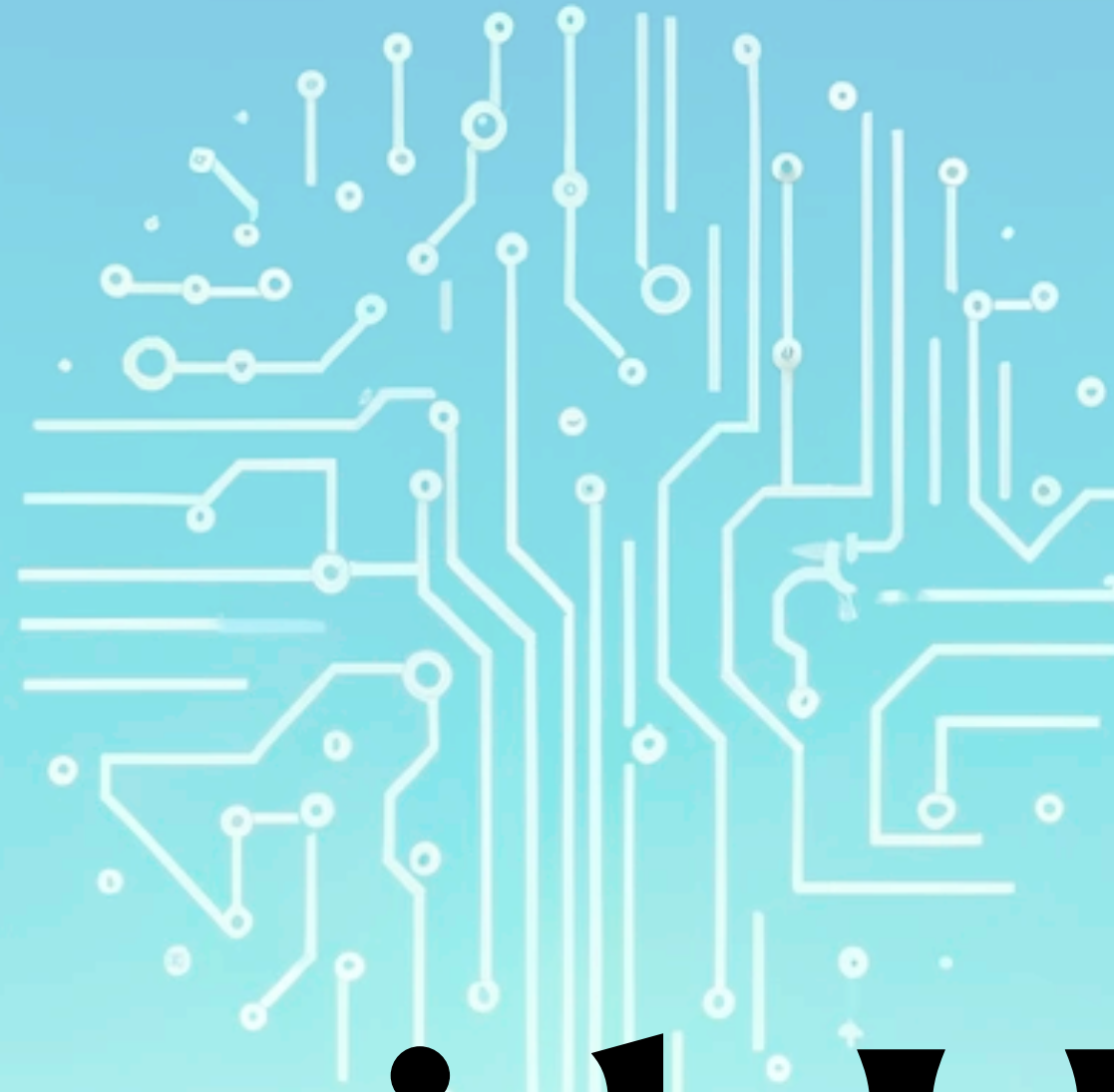




Hacking with LLM-tech

building content discovery and curation at blaze.email

Alastair Rushworth, 25th January 2024

The plan for this evening

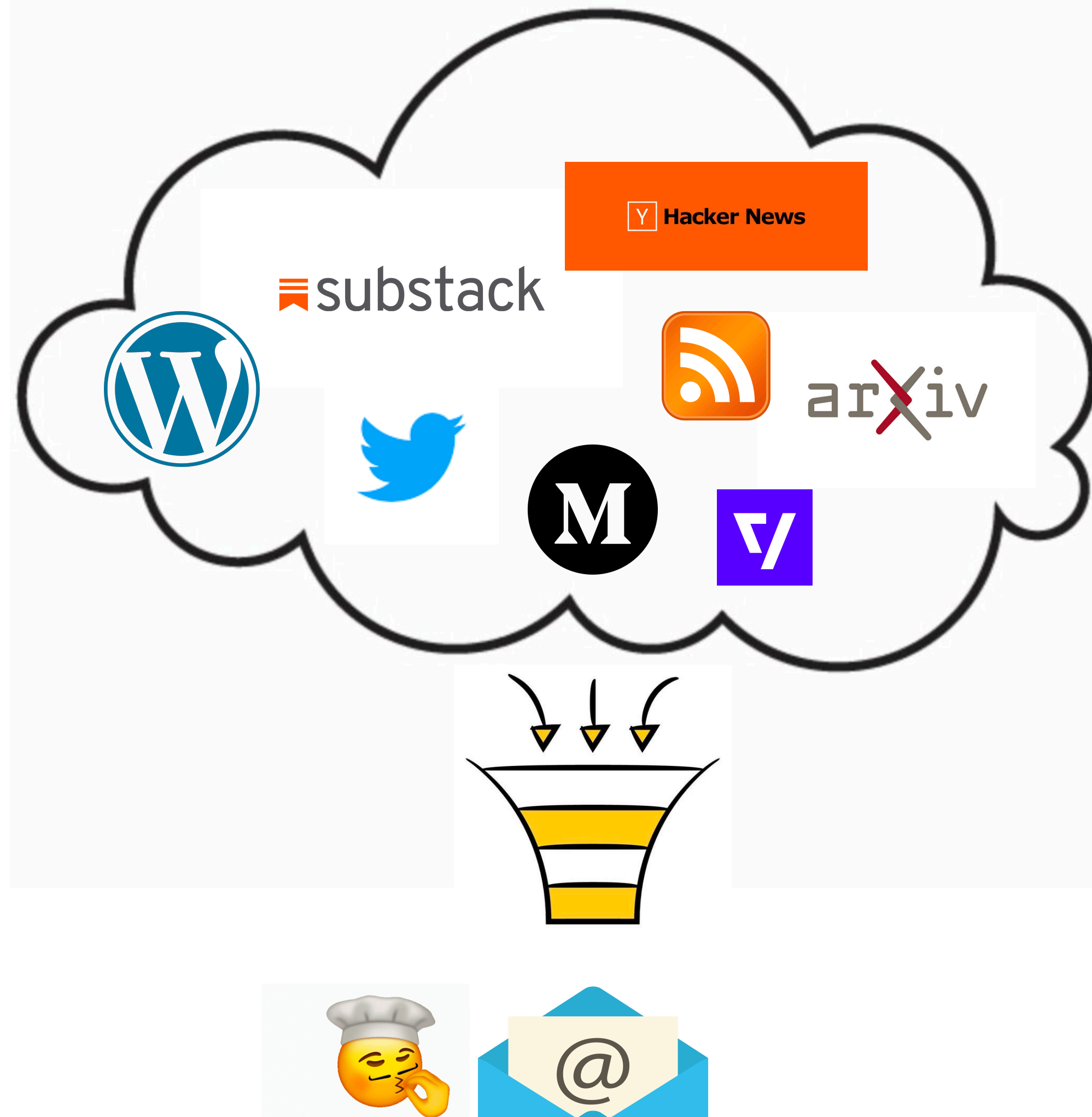
Describe how blaze.email & underlying tech works

Hopefully convey that LLM tech is

- Super exciting and cool
- Very empowering to build with
- Easier to use than you might think

How I got into LLM tech

Blaze: an automated 'mix-tape' of blogs and news for technical professionals



Generative AI

Blaze team, December 5th, 2023

Guides and tutorials

[Large Language Model Course \(github.com, 2023-12-01\)](#). A course on Large Language Models (LLMs) with roadmaps and Colab notebooks

[Extract phone call insights with LLMs in Python \(assemblyai.com, 2023-11-30\)](#). Learn how to automatically extract insights from customer calls with Large Language Models (LLMs) and Python. Extract a summary, action items, and contact information from customer calls using AssemblyAI's Python SDK and LeMUR

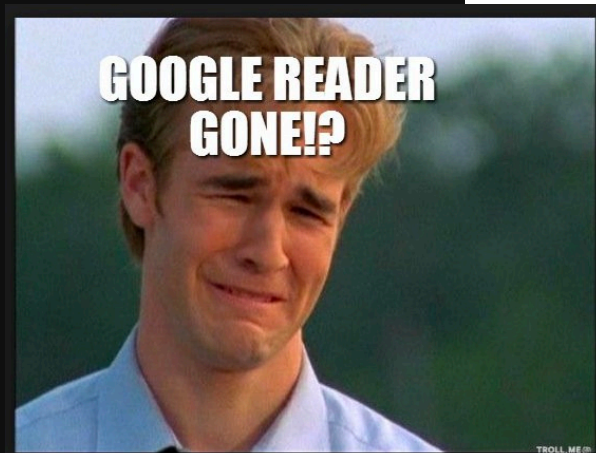
[Comet Launches Course on Building with LLMs \(insidebigdata.com, 2023-11-29\)](#). Comet, a leading MLOps platform, offers a free course on building real-world applications with Large Language Models (LLMs), taught by co-founder of DAIR.AI, Elvis Saravia. The course covers practical examples and use cases, latest trends in LLMs, and how to operationalize them

[How to tackle unreliability of coding assistants \(martinfowler.com, 2023-11-29\)](#). Exploring the dimensions and usage of coding assistants in the context of generative AI and LLMs, and discussing their usefulness and limitations in coding tasks

[Using Honeycomb for LLM Application Development \(honeycomb.io, 2023-11-28\)](#). Using distributed tracing and generative AI with Honeycomb to fine-tune Large Language Models (LLMs) in production for more accurate results

Journal articles

[A Survey on Large Language Model \(LLM\) Security and Privacy: The Good, the Bad, and the Ugly \(arxiv.cs, 2023-12-04\)](#). Exploring the impact of Large Language Models (LLMs) on security and privacy, including their positive contributions, potential risks, and vulnerabilities



A protocol that allows readers to subscribe to updates to a website or blog.

Usually an XML file located somewhere on a website, that can be read by an RSS reader app (like Feedly)

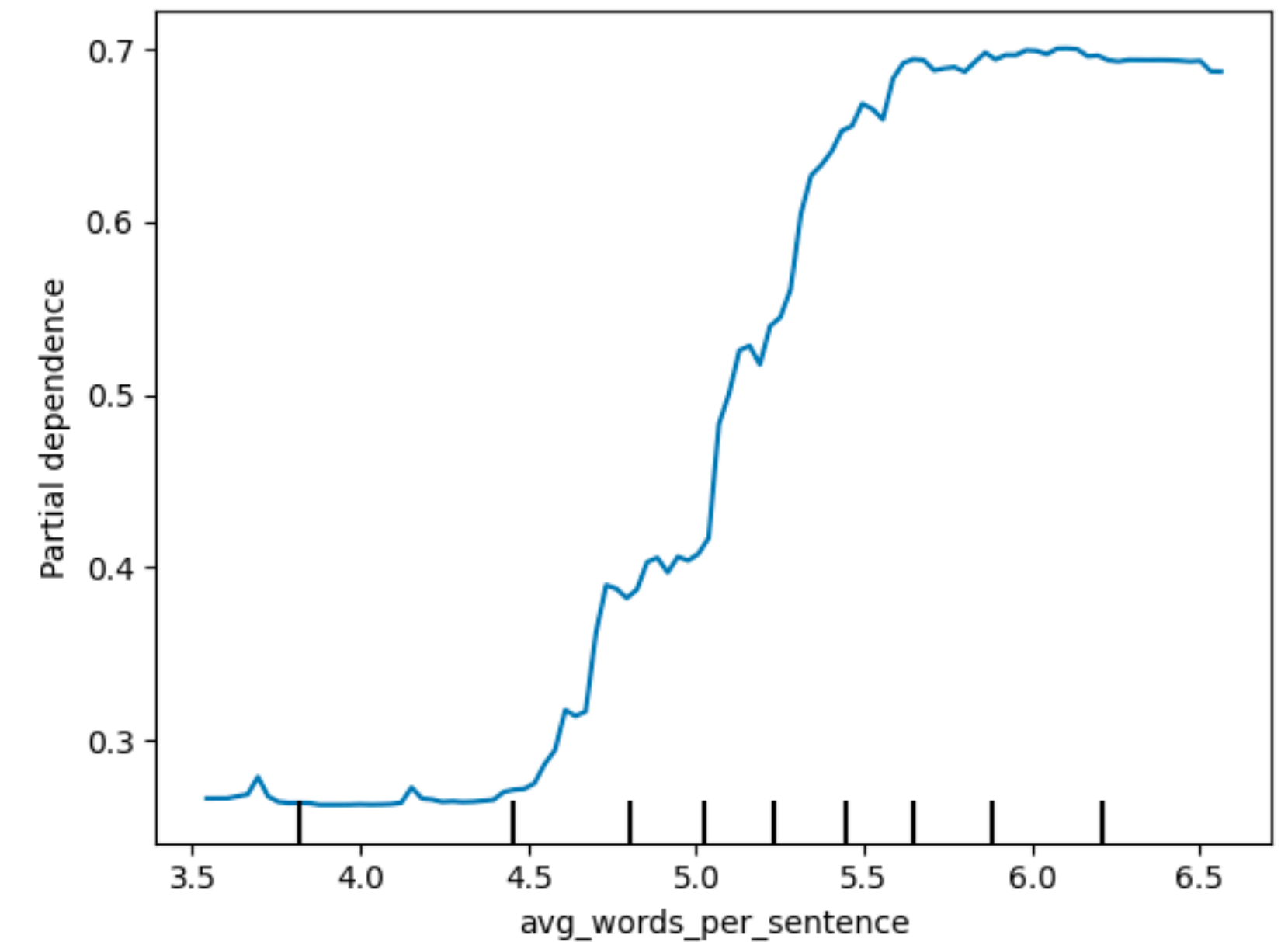
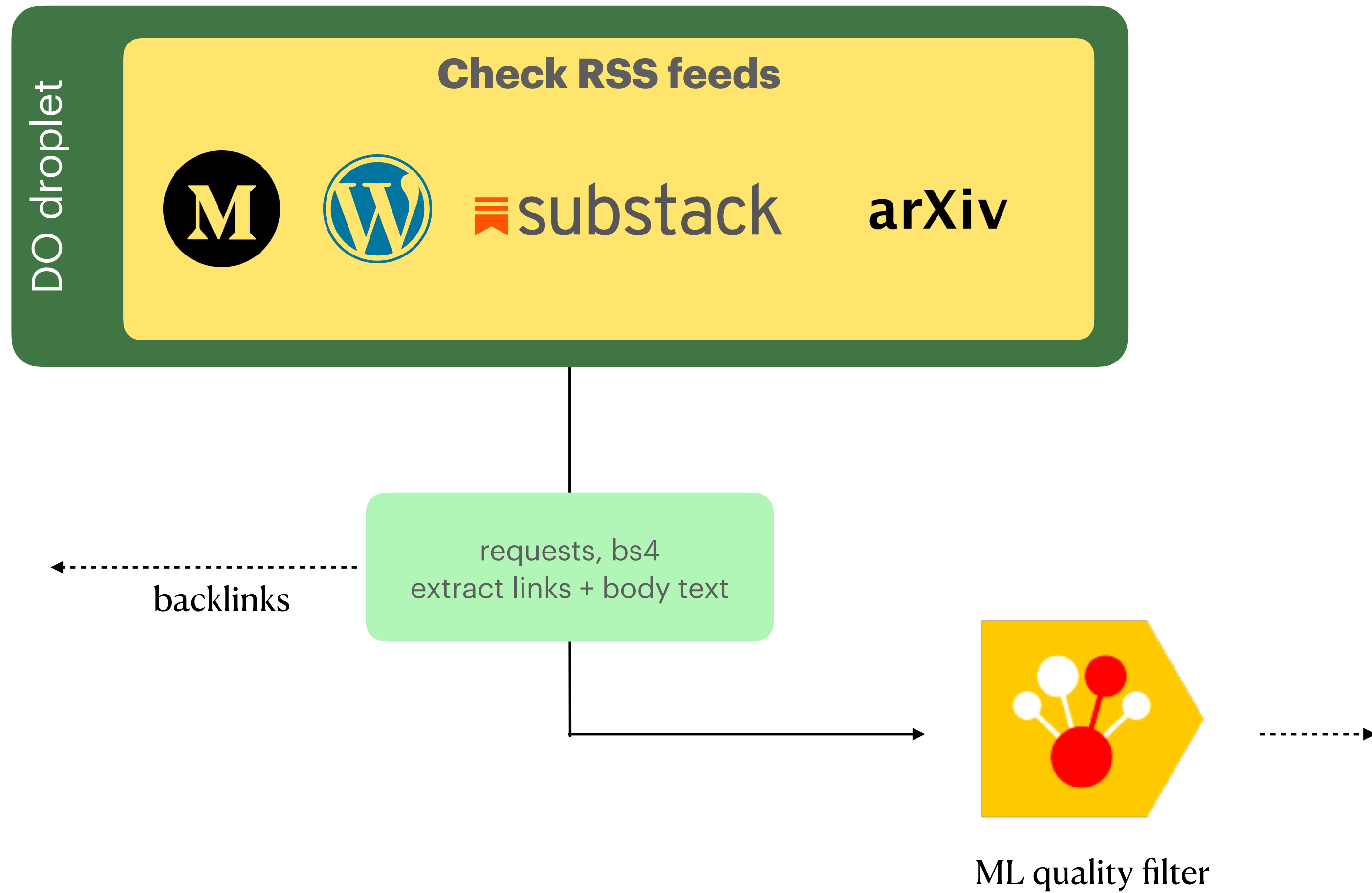
Removes a lot of the need for scraping, which is a time saver for us & good for the host site.

Make it easy(er) to automate fetching new content, provided we have the feed URL.

Blaze tracks ~20k tech-related RSS feeds

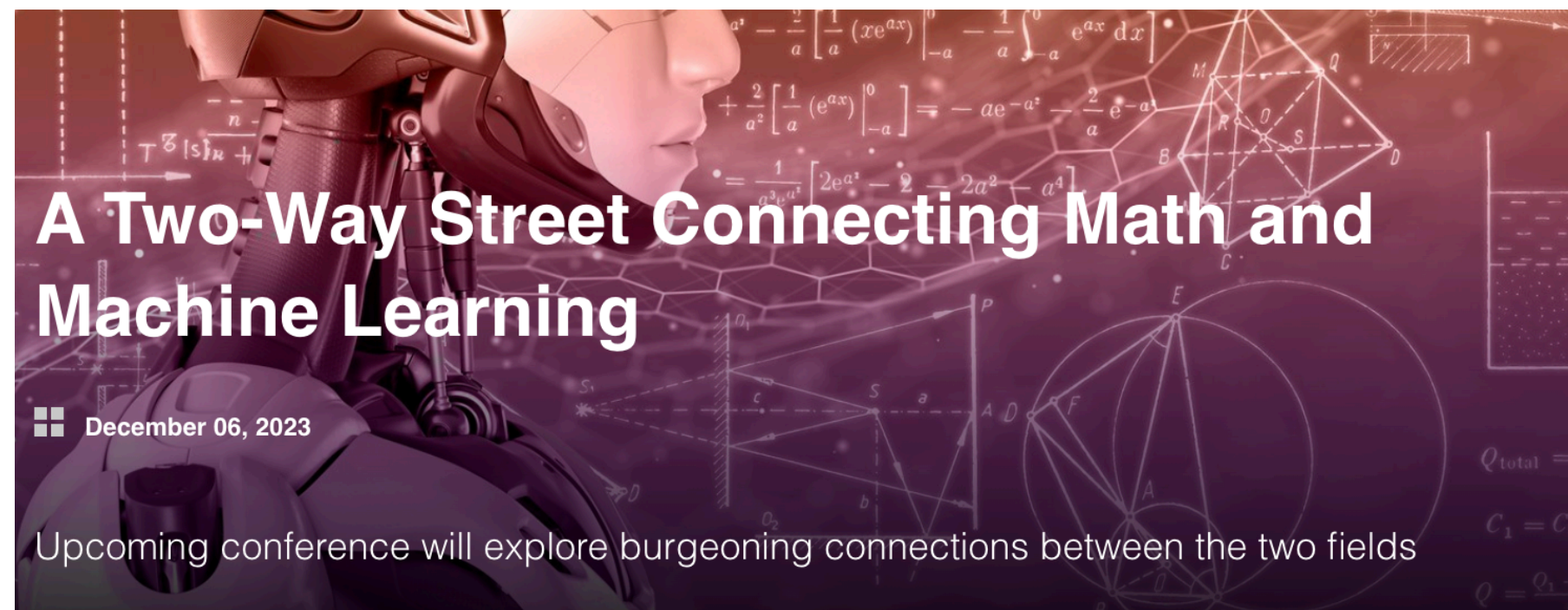


Gathering articles with RSS





Content summarisation

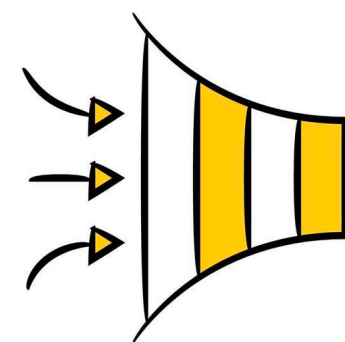


Traditionally, mathematicians jot down their formulas using paper and pencil, seeking out what they call pure and elegant solutions. In the 1970s, they hesitantly began turning to computers to assist with some of their problems. Decades later, computers are often used to crack the hardest math puzzles. Now, in a similar vein, some mathematicians are turning to machine learning tools to aid in their numerical pursuits.

"Mathematicians are beginning to embrace machine learning," says Sergei Gukov, the John D. MacArthur Professor of Theoretical Physics and Mathematics at Caltech, who put together the upcoming **Mathematics and Machine Learning 2023** conference, which takes place at Caltech December 10–13.

"There are some mathematicians who may still be skeptical about using the tools," Gukov says. "The tools are mischievous and not as pure as using paper and pencil, but they work."

<https://www.caltech.edu/about/news/a-two-way-street-connecting-math-and-machine-learning>



Content attributes

language

date

topics

keywords

article type

etc



Content summarisation prompt

```
please return the following pieces of information:
```

- **[contenttype]**: one or two lowercase words describing the type of content on the page. Examples: "podcast", "tutorial"
- **[lang]**: The 2 letter ISO 639-1 code for the content language.
- **[subtitle]**: Write a 35-word subtitle that expands on the title `''' + title + '''` using specific details from the text above
- **[entities]**: A comma separated list of up to 10 of the most relevant entities, technical terms or concepts in the article.
- **[topics]**: A comma separated list of 5 job titles or professions most likely to take an interest in the content.

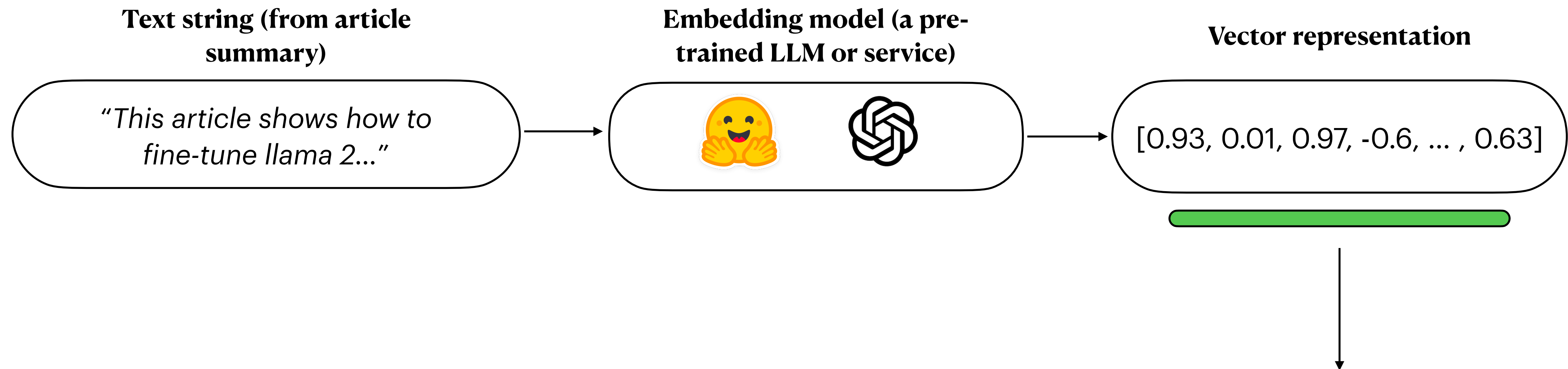
```
return the results as a python dict with the keys "contenttype", "lang", "subtitle", "entities", "topics". Please only use double quotes (") to quote the keys and values in the result to avoid errors with apostrophes. Please ensure empty values are returned as empty strings.
```

Call python's **eval** on the generated text

- returns a **dict** object we can associated with the url + source text
- surprisingly reliable, failure rate is <0.1%: just retry if **eval** throws an error
- use a similar, but separate call to get an abstract length **summary** (up to 200 words)



Embed the content



📖 Recommended reading

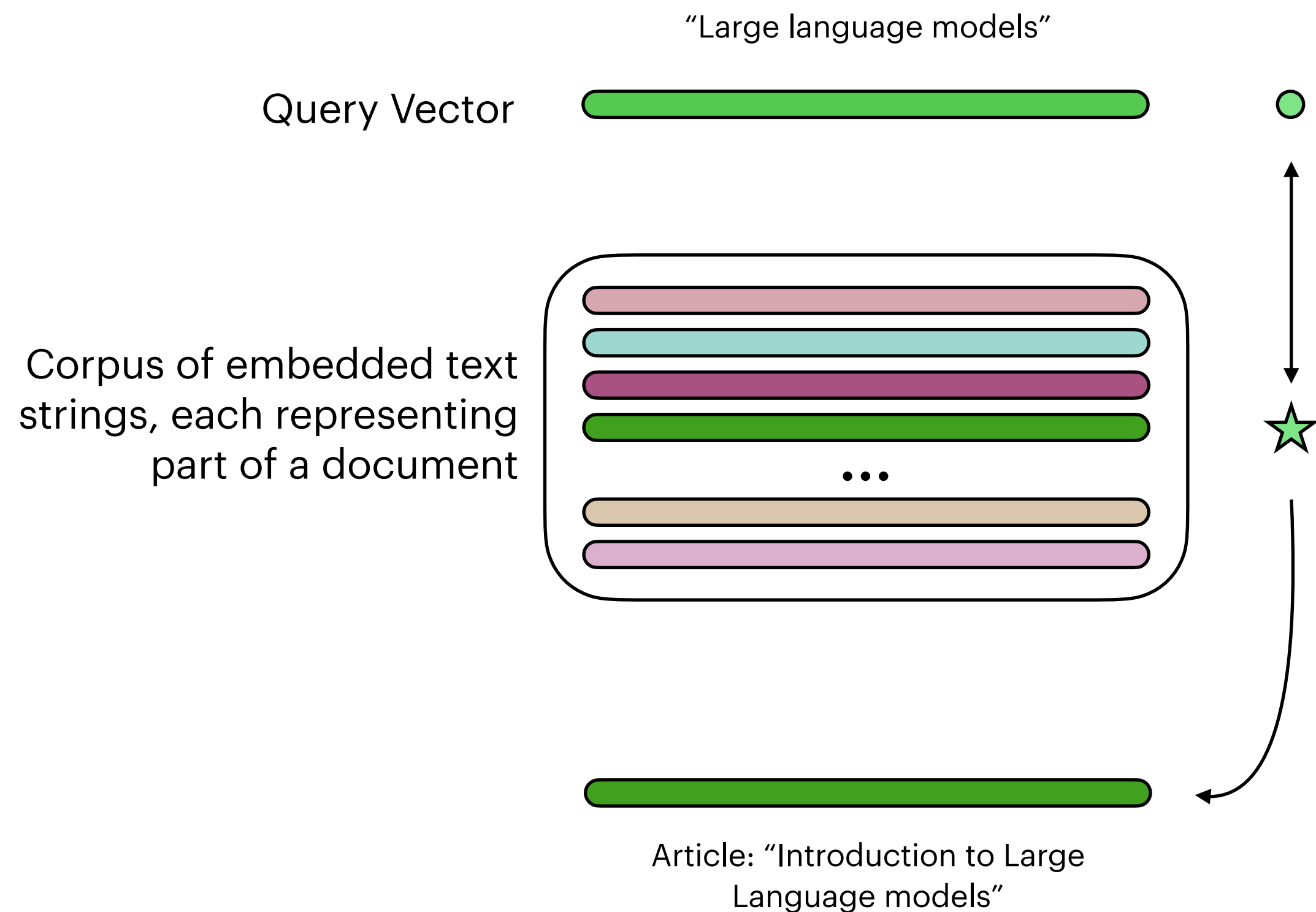
- https://vickiboykis.com/what_are_embeddings/
- <https://simonwillison.net/2023/Oct/23/embeddings/>

Vector used to query a vector database, like Pinecone (or Weaviate)





Querying a vector DB



- **Retrieve** the vectors (and associated text) that are 'close' to the query vector.
- Measure closeness using **Euclidean distance** or **cosine-similarity**.
- Very fast, hybrid & approximate search - not $O(n^2)$!



Relevance, quality and conciseness

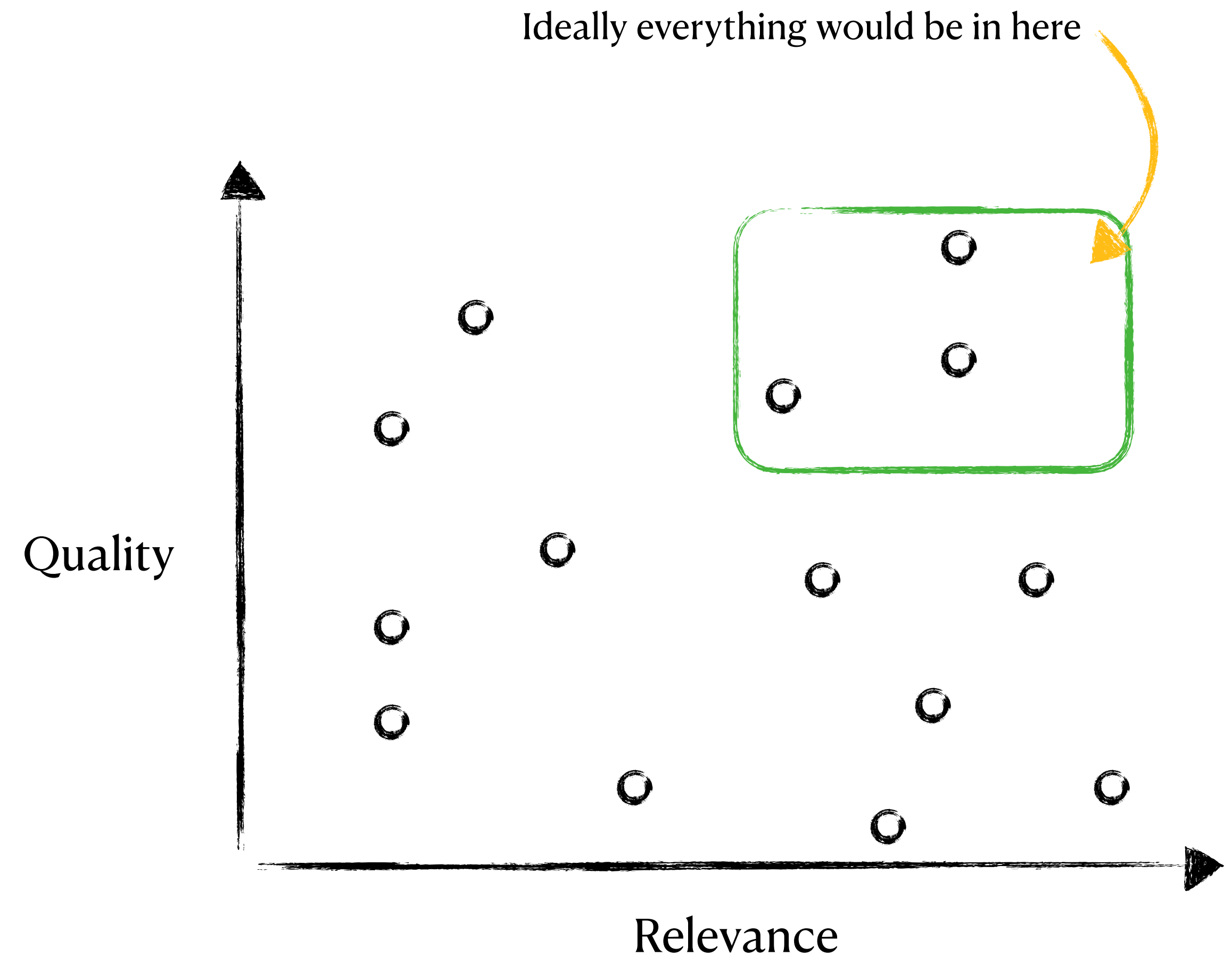
Highly relevant documents can be

- Low in quality (eg. poorly written, contain wrong or outdated information, not from a credible source)
- Not concise (low volume of factual information vs number of tokens)

Setting up good retrieval depends on strong data quality

- Recent and task specific
- Use canonical sources if they exist

THE biggest problem when I was building Blaze





How items look in the vector db

ID

https://www.caltech.edu/about/news/a-two-way-street-connecting-math-and-machine-learning

Values

-0.00669013895,-0.00068163214,0.0162437093,-0.0173521359,0.002418

⊕

Add Sparse Data

Key

Value

rss

https://www.caltech.edu/about/news/rss/

Aa

Key

Value

rsstype

blog

Aa

Key

Value

score

0.985

123

Key

Value

subtitle

Mathematicians and machine learning experts come together to explore the connections between the two fields and how they can

[]

Key

Value

title

A Two-Way Street Connecting Math and Machine Learning

Aa

Key

Value

topics

mathematicians, machine learning experts, data scientists, computer scientists, AI researchers

[]

Key

Value

unix_time

1701900730

123

Unique ID (str)

Text embedding (array)

Metadata (upsert via dict)



Metadata queries

```
from pinecone import Pinecone

pc = Pinecone(api_key="XXX_YYY_ZZZ")
index = pc.Index("pinecone-index")

index.query(
    vector=[0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, ...],
    filter={
        "score": {"$gt": 0.9},
        "unix_time": {"gt": 1705913604},
        "rsstype": {"$eq": "blog"}
    },
    top_k=10, # number of results to return
    include_metadata=True # returns metadata with each result
)
```

Query embedding

you have to get this from your query string + embedding model before constructing this query

Metadata filters

The metadata filters can be combined with AND and OR:

- `$eq` - Equal to (*number, string, boolean*)
- `$ne` - Not equal to (*number, string, boolean*)
- `$gt` - Greater than (*number*)
- `$gte` - Greater than or equal to (*number*)
- `$lt` - Less than (*number*)
- `$lte` - Less than or equal to (*number*)
- `$in` - In array (*string or number*)
- `$nin` - Not in array (*string or number*)

Source: <https://docs.pinecone.io/docs/metadata-filtering>

Blaze search



Search [Subscribe](#) [About](#)

large language model evaluation benchmarks since:last_week lang:en 66/200

academic



25 results (0.00 seconds)

[DevEval: Evaluating Code Generation in Practical Software Projects](#)

12-01-2024 (0.866) <http://arxiv.org/abs/2401.06401v1>

New Benchmark DevEval Evaluates Code Generation in Large Language Models for Practical Software Projects

→ [More like this](#)

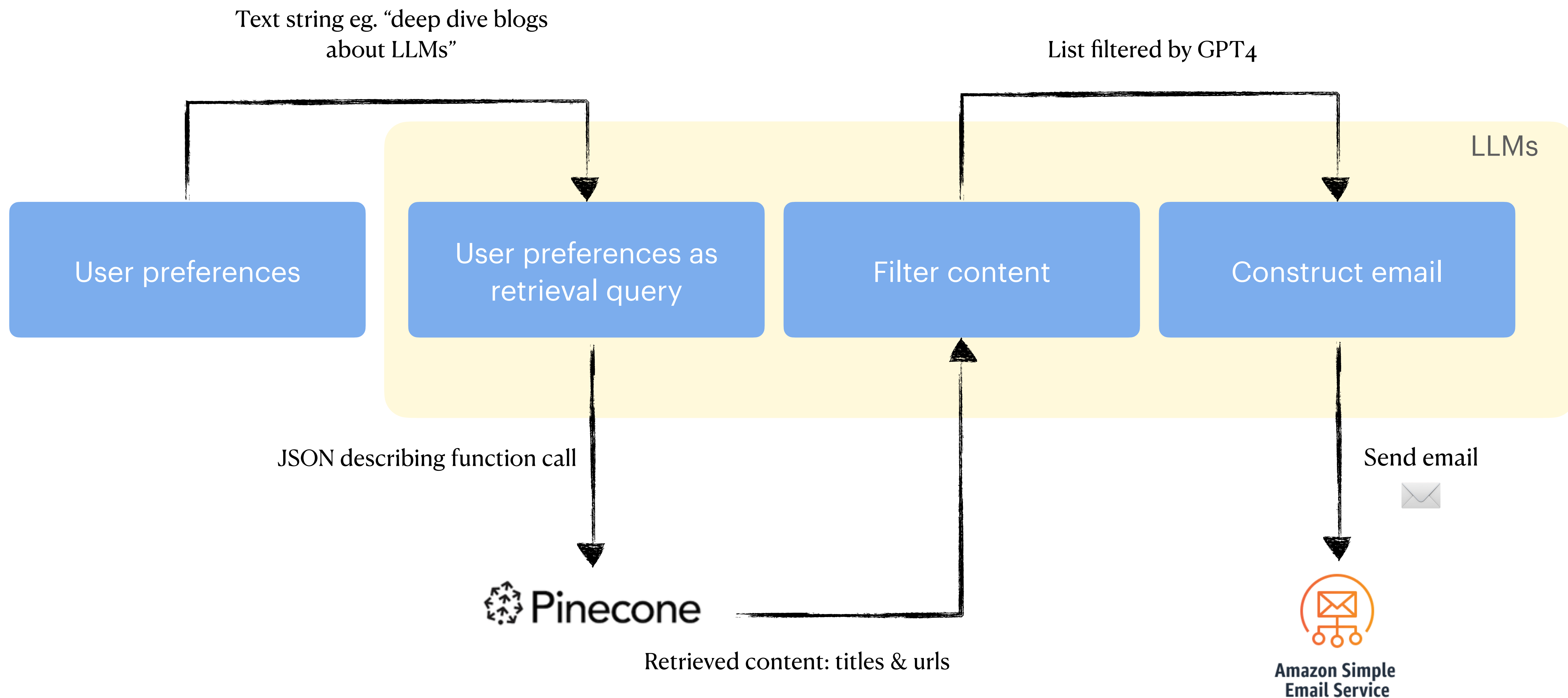
[PersianMind: A Cross-Lingual Persian-English Large Language Model](#)

12-01-2024 (0.863) <http://arxiv.org/abs/2401.06466v1>

PersianMind: A Cross-Lingual Persian-English Large Language Model that Rivals GPT-3.5-turbo in Persian Language Proficiency by Expanding LLaMa2's Vocabulary and Training on a Persian Dataset

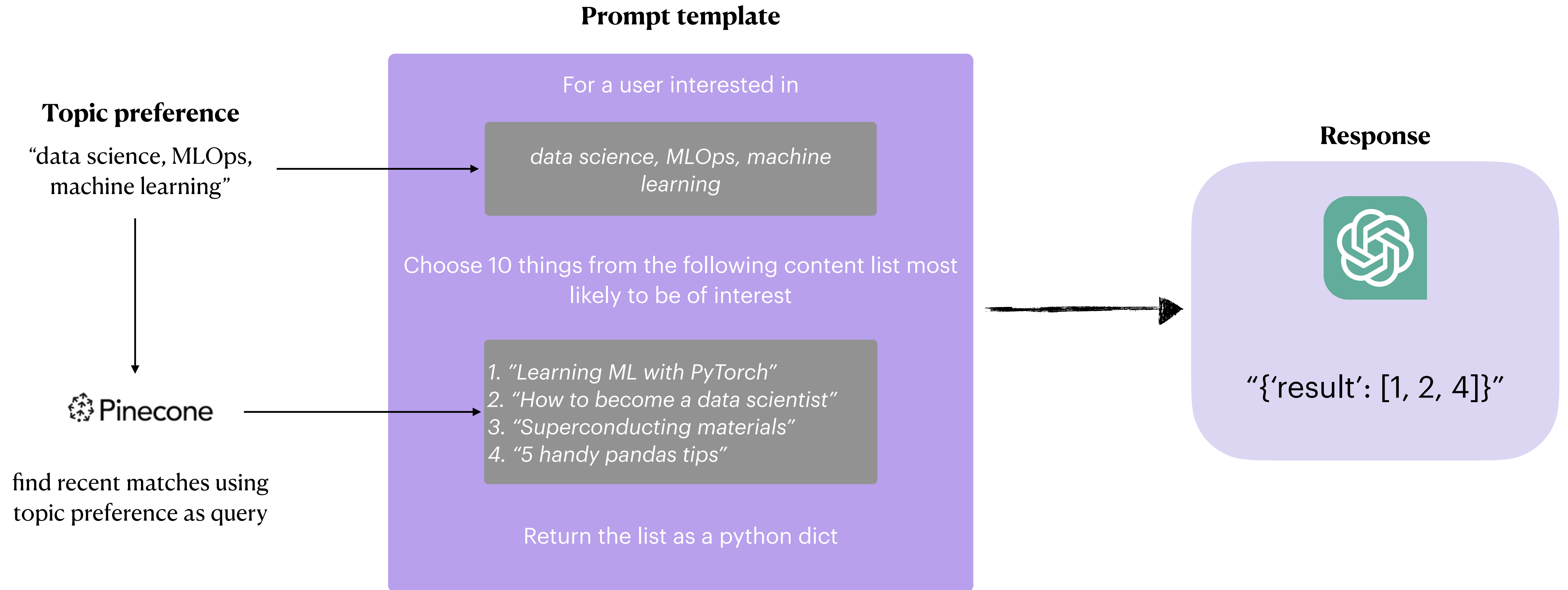


Blaze email



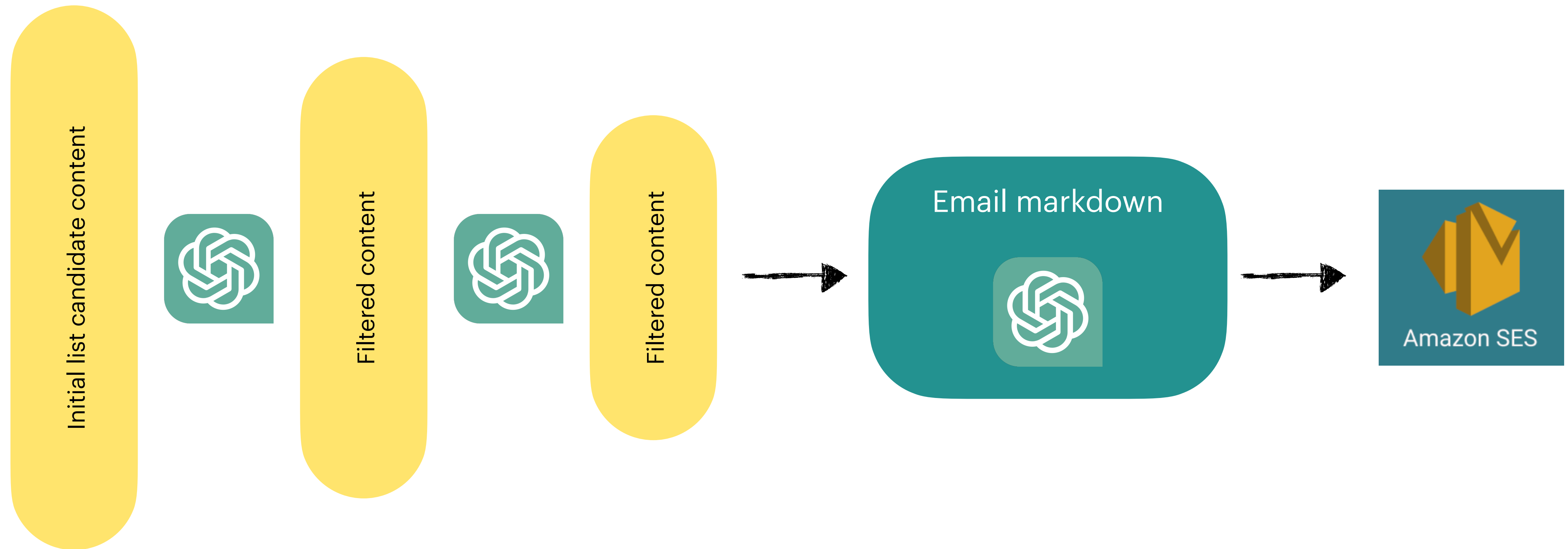


RAG: content curation version



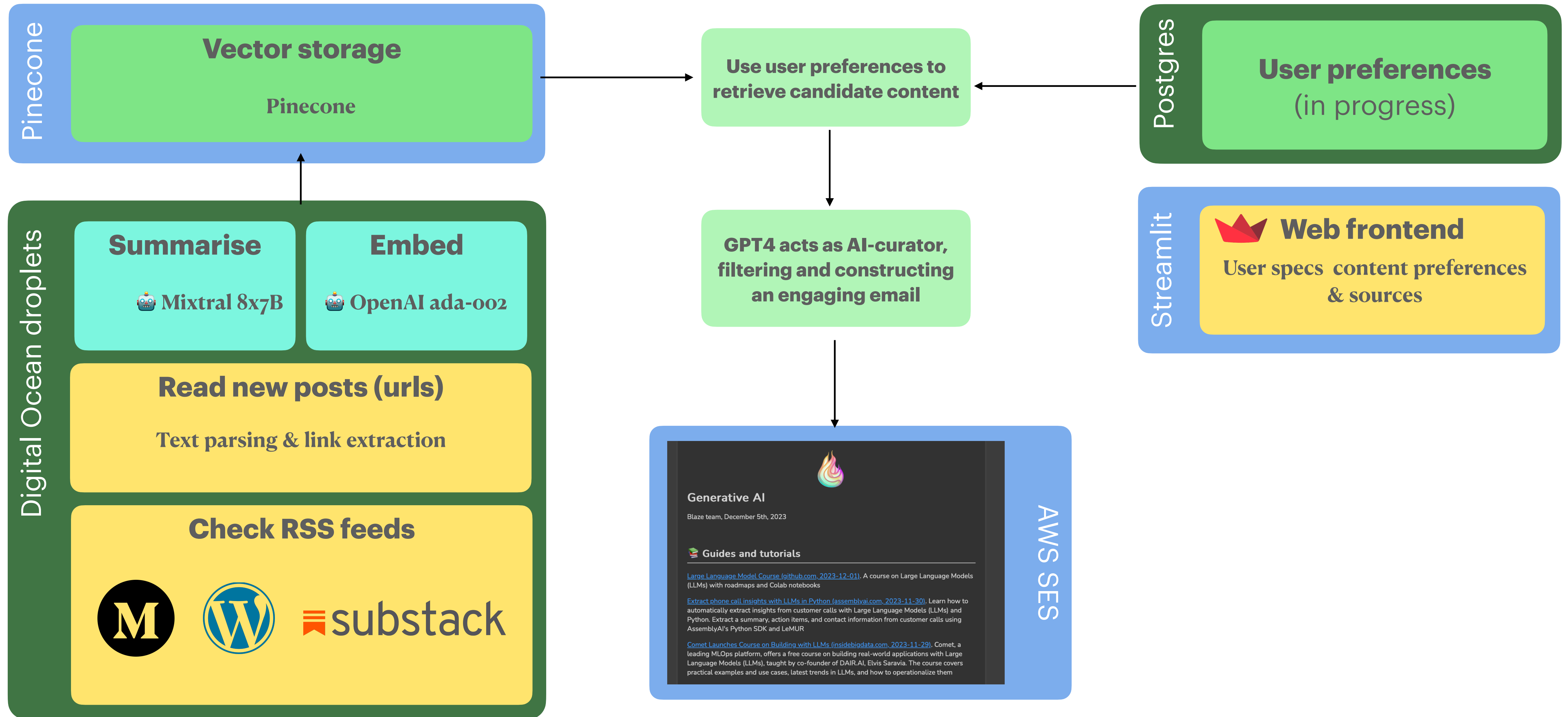


RAG: content curation version





blaze.email architecture





Thanks!



Source: JupyterMeowbooks

(<https://twitter.com/untitledoipynb>)